



Evidence and Argument Tables

Haul truck and light vehicle collision, Bindarra Gold Mine

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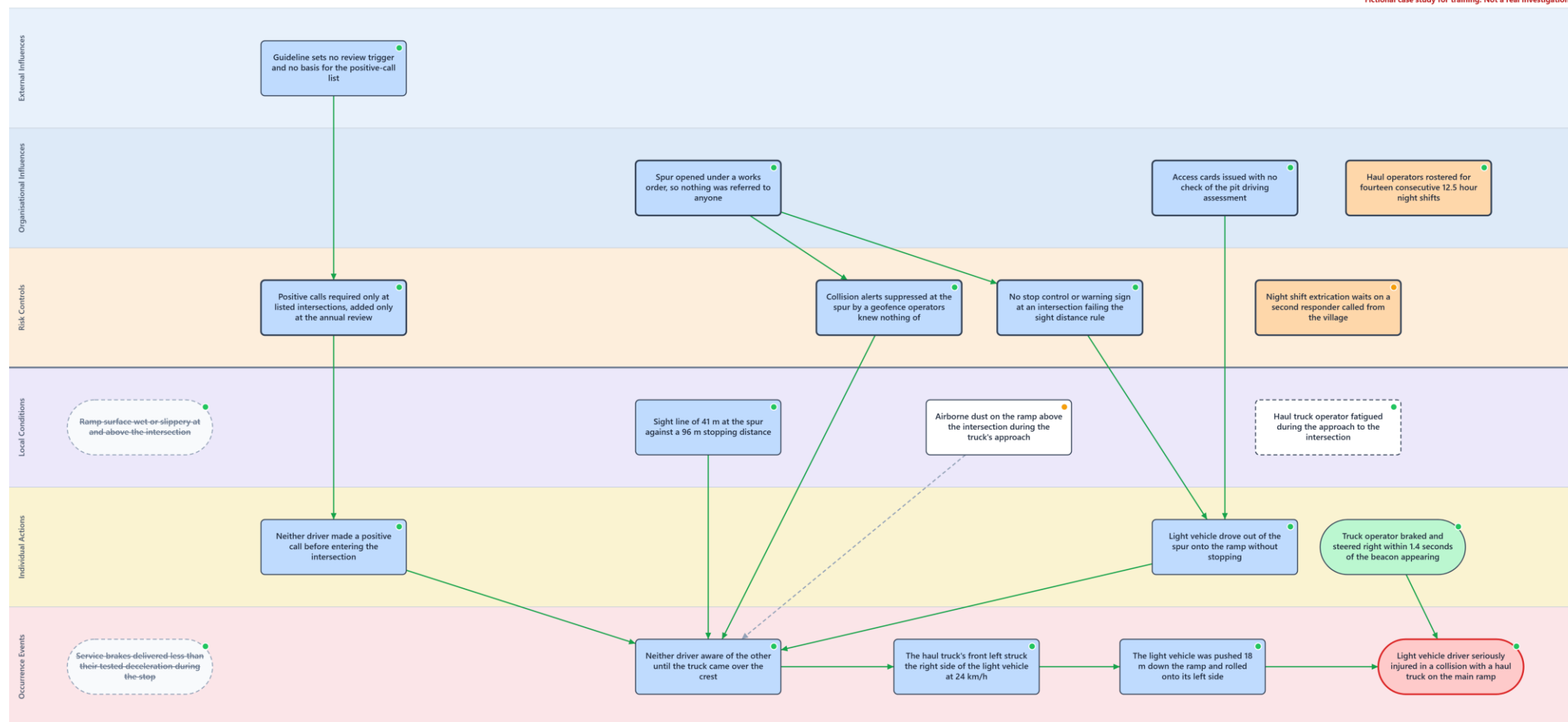
INVESTIGATION WORKFLOW SUITE

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AcciMap

Haul truck and light vehicle collision, Bindarra Gold Mine

Rendered 08/10/2026, 4:49:00 pm
Fictional case study for training. Not a real investigation.

How to read these tables

Each table is one object from the analysis map. An object is a factor: an event (something that happened) or a condition (a state of affairs that existed at the time). Each factor is worded as a proposition, a statement that is either true or not, which the three tests then examine.

The tests run in order. Existence asks whether the factor existed or occurred. Influence asks whether it influenced this occurrence. Importance asks whether it is worth addressing regardless. A later test only means something once the earlier one is settled: Influence is tested once Existence has passed, and Importance where Influence failed. A test not yet reached reads Not tested.

The shaded row opens each test with its result (Pass, Fail, Inconclusive or Not tested) and the confidence behind it. Confidence follows the probability scale: virtually certain, extremely likely, very likely, likely, more likely than not, about as likely as not, unlikely, very unlikely, extremely unlikely and exceptionally unlikely, with the percentage in brackets. High confidence passes the test, low confidence fails it, and the middle is Inconclusive.

Under the result, Argument For sets out the case that the factor, as worded, is true for that test, and Argument Against the case that it is not. Both are always read against the factor as worded, never against the result, so a failed test still shows its case for. Reasoning says which side carried and why.

Nil means nothing was recorded on that side. A test argued both ways shows a case on each side, and the reasoning says which one carried. A test with nothing recorded shows only its result row: no argument was written for it, so its result is not supported by this record.

External Influences

Guideline sets no review trigger and no basis for the positive-call list

Existence:	Pass, virtually certain (100%)
Argument For: MSG-14 clause 3.2 requires the traffic management plan to be reviewed within twelve months and after any significant change to the road network, and defines neither the term nor an example of one. Clause 5.1 requires the operator to nominate the intersections and interaction zones at which positive communication is required, and sets no minimum and no basis for the nomination. Clause 3.5 provides that the plan is a site document, not approved by and not submitted to the Inspectorate except on request after an occurrence. Bindarra's plan met the guideline on its face on the day of the occurrence: it had been reviewed inside twelve months and it nominated nine interaction zones. The mine was not in breach. Clause 5.4, which offers five alternative means of making a light vehicle detectable, any one of which satisfies it, is context rather than part of this factor. Argument Against: Nil Reasoning: The guideline's text and the regulation it is made under are documentary and not in dispute.	
Influence:	Pass, likely (70%)
Argument For: Had the guideline defined a significant change and set a basis for choosing positive-call intersections, a new intersection on a haul road would have had a defined route onto the plan's call list between annual reviews, and the eleven Stage 4 roads never referred to the plan's owner would have had somewhere to land. As written, both terms are left to the same site document, reviewed once a year, with no outside eye on either. Argument Against: The guideline is not what stopped anybody. A site that wanted an out-of-cycle review could hold one under its own clause 9.1 at any time, and MSG-14 does not prevent a mine from listing every intersection it has. Bindarra did not fail to act because the guideline was vague; it failed to act because nothing inside the mine ever put the question in front of the plan's owner. On that reading the external clause is background and the organisational failure is the whole of it. Reasoning: Argued, and accepted at 70% on the balance, at the standard of proof and no higher. The guideline bears on one path, the listing path: a defined trigger would have required the plan to be reopened when a road was added, and a stated basis for nomination would have given an intersection that fails the sight distance rule a claim on the list. The Against is real, because the mine could have done both on its own, and the 70% says only that the guideline made doing nothing the default. The report should say it is held at the comfort line rather than presenting it as settled.	
Importance:	Not tested

Organisational Influences

Spur opened under a works order, so nothing was referred to anyone

Existence:	Pass, virtually certain (100%)
Argument For: ORR-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design where no new plant is introduced. WO-26-4471 cut 240 m of access to bore DW-14 at RL 312, classified the work as routine earthworks within the approved Stage 4 design, recorded management of change as not required under clause 5.3, and left its handover field reading N/A. The three referrals that a management of change record would have triggered at clause 5.4 are the traffic management plan owner, the survey department for the fleet management system road network, and the contractor's technical services group for the collision awareness system geofences. None of the three was made. The in-cab road network published on 24/07/2026 was still current on the night and shows the main ramp with no intersection at RL 312, and the S4-LOAD suppression geofence, carried up the ramp on 09/06/2026, already covered the place where the spur joined it. Of the fourteen new roads cut in Stage 4 since the last plan review, the three raised under a management of change record were referred to the plan's owner and none of the eleven raised under a works order was. Argument Against: Nil Reasoning: The procedure, the works order and the road register are documentary and agree with each other.	
Influence:	Pass, very likely (90%)
Argument For: A works order built an intersection into a haul road and the classification on its front page removed every consequence of having done so. Eleven roads have been built the same way since the last review. Argument Against: Nil Reasoning: Had a management of change been raised, clause 5.4 would have sent the new intersection to the plan's owner, with a survey showing it failing the clause 6.2 sight distance rule and so needing a stop control and the clause 6.4 signs, and to the technical services group, who configure the geofence the intersection was cut inside. Both referrals are standing parts of the procedure, and the three roads raised under one since the last review were referred. The 10% allows that a referral is not a control: the plan's owner might have left the spur to the annual review, and no out-of-cycle review has been held in the plan's eleven years.	
Importance:	Not tested

Access cards issued with no check of the pit driving assessment

Existence:	Pass, virtually certain (100%)
Argument For: ORR-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement, and Hallorn Survey & Geotechnical has held one since 2023. Clause 7.4 states that equivalence does not extend to the Bindarra site-specific supplement or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive in an operating pit and shall not be substituted. Clause 7.5 makes the engaging department responsible for verifying 7.4 before a card is issued. The gatehouse issues cards against the recognition agreement alone, holds no record of either item, and is not asked for one; nothing in the access card system distinguishes a person who may drive in an operating pit from a person who may not. The driver's card was issued on 09/08/2026 after a 20 minute gatehouse briefing. His training file records the Hallorn online module of 24/07/2026 and no pit driving assessment; he had never heard of one and had never seen BIN-TMP-004. Whether the site supplement was issued is not yet known, because Hallorn's own records for it are still outstanding. Argument Against: Nil Reasoning: The standard, the gatehouse procedure, the card register and the training file are documentary and agree: no assessment was asked for and none was done.	
Influence:	Pass, likely (75%)
Argument For: Had the gatehouse checked for the pit driving assessment, he would not have been carded to drive in an operating pit until he had done it, and the assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice: giving way to haul trucks, approaching an intersection he cannot see up, and when calls are made. He said that if he had been told to call, he would have called. Argument Against: He was already applying the rule the assessment teaches first, that light vehicles give way to heavy vehicles at all times, and he slowed and looked before he went. The plan requires no call at an unlisted intersection, so the assessment would not have taught him to call at RL 312, and his 'I would have called' cannot be stretched that far. Reasoning: Argued, and accepted at 75%. What the assessment adds over what he did is how to approach a mouth he cannot see out of: stop, and look again, rather than slow. On the recorded times a stop at the mouth would very likely have shown him the truck. Held below the other influence findings because the record holds no copy of the assessment's content, and because he was already applying its first rule.	
Importance:	Not tested

Haul operators rostered for fourteen consecutive 12.5 hour night shifts

Existence:	Pass, virtually certain (100%)
Argument For: The contractor's Stage 4 haul roster is 14 nights on and 7 off, 18:00 to 06:30 local, a 12 hour 30 minute shift with a 30 minute crib, inside a fatigue management standard that permits up to 14 consecutive night shifts with a minimum 10 hour break. The haul truck operator was on night 10 of 14, and the collision at 02:47 was 8 hours 47 minutes into the shift, in its ninth hour and inside the window of lowest circadian alertness. Every limit in the standard was met, and the previous nine shifts were worked in sequence with none extended beyond 12 hours 40 minutes. The night shift supervisor works the same pattern as the crews he supervises. Argument Against: Nil Reasoning: The roster, the site access records and the contractor's fatigue standard are documentary.	
Influence:	Fail, unlikely (25%)
Argument For: Had the roster been shorter, the operator would not have been at the wheel at 02:47 on a tenth consecutive night, which is where degraded detection and slowed response are expected, on a task, picking out a hazard in an unlit pit, that fatigue degrades first. Argument Against: The download gives 1.4 seconds from the beacon coming into view to a brake pressure rise, inside the 1.5 seconds the plan's own stopping distance calculation assumes, and the braking and steering that followed were correct and immediate. The arithmetic settles it: braking at the recorded 0.78 m/s² from 53 m, a response time of zero still ends in a collision, at about 17 km/h instead of 24. The 1.4 seconds cost about 7 km/h of impact speed, not the collision, and there is no decision or omission on the download for a roster to have worked through. Reasoning: Failed on the evidence, not dismissed. The one mechanism the roster could have worked through here, a slow or absent response, is contradicted by the recorded data, and the sight line removes the margin a faster response would have needed.	
Importance:	Pass, likely (85%)
Argument For: It failed Influence for this occurrence and remains a finding worth acting on. The Stage 4 roster puts every haul truck operator at the wheel of a 220 tonne vehicle in the small hours of a tenth, twelfth and fourteenth consecutive night on every swing, where the defence against an unlit hazard at an unlisted intersection is the operator's own detection and response. Nothing about that changes because it did not bite here: at an intersection with a longer sight line than RL 312, a slower response is the difference between stopping and not. Argument Against: Nil Reasoning: A standing exposure built into the roster rather than a feature of one night.	

Risk Controls

Positive calls required only at listed intersections, added only at the annual review

Existence:	Pass, virtually certain (100%)
Argument For: BIN-TMP-004 clause 4.3.1 requires a positive call on the pit channel before entering any intersection listed at Appendix B, and 4.3.2 requires no call anywhere else. Appendix B lists nine intersections. Its own note provides that the appendix is reviewed and reissued at the annual review, and that an intersection is added where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication. Schedule 1 is dated 12/11/2025 and is reissued at the annual review. No provision anywhere in the plan allows an intersection to be added between reviews, and no provision requires a new road to be referred to the plan's owner when it is cut. The RL 312 spur is on neither document. Eleven of the fourteen new Stage 4 roads since the last review were never referred, and no out-of-cycle review has been held in the plan's eleven year history. The pit channel recording shows the practice: HT-407 called, and was acknowledged, at the RL 360 crossover going up and coming down and at the ROM pad turn-off, the last of them 51 seconds before the collision, and made no call at the unlisted spur. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently: calls are made at the nine intersections on the crib room board and nowhere else. Argument Against: Nil Reasoning: The plan's text, its appendix and its review history are documentary.	
Influence:	Pass, very likely (90%)
Argument For: This is the mine's own defence against exactly this interaction, and at this intersection it was unarmed by its own terms. The next date on which the spur could have been listed is 12/11/2026, three months after the collision. Argument Against: Nil Reasoning: Had RL 312 been listed, the call would have been made: the operator called at every listed intersection he passed that night, the last of them 51 seconds before the collision, and the channel was open with the light vehicle on it. The 10% allows that a call made and not acknowledged still leaves a driver deciding whether to go.	
Importance:	Not tested

Collision alerts suppressed at the spur by a geofence operators knew nothing of

Existence:	Pass, extremely likely (95%)
Argument For: The collision awareness system on the Stage 4 haul fleet operates at Level 8, operator advisory; Level 9 intervention is available on the fleet and is not enabled at Bindarra. Advisory alerts are raised on a tag entering the 60 m ring above a 10 km/h closing rate, except inside a suppression geofence, where no alert is raised at any range and a proximity event is logged only once the range falls below 25 m. The S4-LOAD geofence was drawn around the loading face on 14/03/2025 to stop nuisance alerts and extended northward on 09/06/2026 when the face advanced, which carried the boundary up the main ramp for the first time, to a point 180 m up-ramp of the RL 312 intersection. The spur, cut from 28/07/2026, therefore joined the ramp inside the suppressed area. The extension was made as a system configuration task by the contractor's technical services group, is not recorded in the traffic management plan, and was not reviewed against it. The log records a proximity event at 02:47:09.2 (18:47:09.2 UTC) at 21.4 m and 32 km/h closing, alert suppressed. In the twelve months before the occurrence HT-407 raised 1,684 advisory alerts, every one outside the geofence and none inside it; across the fleet 68 proximity events inside it were suppressed, nine of them a light vehicle on the main ramp below the boundary, all after 09/06/2026. No training record anywhere on site, for any person, covers the geofences, and the truck operator and the night shift supervisor each said they did not know the system was off on the ramp. Argument Against: Nil Reasoning: The configuration, the event log and the statistics come from the system itself and were downloaded sealed. The 5% is the second half of the proposition, that operators did not know: it rests on two accounts and on the absence of any training record or notice, and no other operator has yet been asked.	
Influence:	Pass, likely (70%)
Argument For: The operator learned over fourteen months that a quiet system on the ramp meant an empty ramp. For the first ten of them that was true, because the ramp was outside the geofence and there was nothing there to alert on. From June the system was silent there whatever was on the road, and nobody told him. He wrote in his own report, before he was asked, that he 'took that to mean the pit was empty when it was quiet', and he came over the crest carrying that expectation with nothing else on the ramp to tell him there was an intersection below it. Argument Against: The alert itself would have bought nothing. Unsuppressed, it would have sounded when the light vehicle's tag entered the 60 m ring, at about 02:47:06.5, half a second after the beacon came into view and about a second before the brakes went on, so it would have told the operator what he was already reacting to. The system is advisory only and carries no intervention. Nor is the expectation shown to have changed anything he did: he was descending at the speed every operator uses on that ramp, and his own remedy was a sign ('I'd have been at twenty'), not the system. Reasoning: Argued, and accepted at 70%, at the standard of proof and no higher; re-argued down from 75% once the alert timing was corrected. The mechanism is the expectation: an operator who believed the system would sound for a light vehicle on the ramp had one less reason to think anyone could be there, on a ramp that showed no intersection on his screen and none on the road. That is a real effect and an unmeasurable one, and the report should carry both readings.	
Importance:	Not tested

No stop control or warning sign at an intersection failing the sight distance rule

Existence:	Pass, virtually certain (100%)
Argument For: BIN-TMP-004 clause 6.2 requires the available sight distance at every intersection, listed or not, to be not less than the stopping distance of the largest vehicle at the posted speed and in no case less than 120 m, and where that cannot be achieved requires the intersection to be controlled by a stop control, signalised control or traffic controller. The survey gives 41 m against a stopping distance of 96 m, so the second limb applied. Clause 6.4 requires an intersection warning sign 50 m in advance on each approach, a give-way control on the minor road and delineation at the mouth, at every intersection shown on the road layout drawing. At the time of the occurrence the RL 312 intersection had no warning sign on either ramp approach, no stop or give-way control at the spur mouth, no delineation and no posted speed. The construction sign at the mouth was removed on 02/08/2026 when the spur was first used, and nothing was put in its place. The hazard report of 04/08/2026 named the missing sign and the missing give-way. Argument Against: Nil Reasoning: The signage examination, the works log and the plan's text agree, and the hazard report describes the same absence seven days earlier.	
Influence:	Pass, likely (75%)
Argument For: A stop control at the spur mouth would have held the light vehicle there while the truck came into his view, and a warning sign on the truck's approach would have told the operator there was an intersection below the crest. The operator said so himself: 'If there'd been a sign on my side of that crest I'd have been at twenty and I'd have been looking for him.' On 04/08 the technician who came out of the same spur in front of a truck reversed back into it and let the truck pass. Argument Against: A stop line is only as good as the look from it. From the mouth the windrow and the grade break give 38 m, about four seconds of a truck at 37 km/h, which is a thin margin even from a standstill. He was already slowing and looking, on a give-way rule he described applying. Whether a sign would have slowed the truck is the operator's own view after the event, and a sign at an intersection that is not on the in-cab map would have been a single cue. Reasoning: Argued, and accepted at 75%. On the recorded times the truck was inside the light vehicle's 38 m of view from about 02:47:07.4, nearly two seconds before he reached the mouth, so a full stop there would very likely have shown it to him and he would have let it go, as the technician did a week earlier. Held below the other influence findings because the margin from a stop at 38 m is thin and the effect of a sign on the truck rests on the operator's own account.	
Importance:	Not tested

Night shift extrication waits on a second responder called from the village

Existence:	Pass, virtually certain (100%)
Argument For: The Bindarra night shift is rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder, who must be present before any extrication is begun, is called in from the accommodation village 11 km from the gatehouse, in the second response vehicle, which is garaged there and carries the hydraulic rescue tool. On the night, dispatch paged the team at 02:48, the first response vehicle entered the pit at 02:51 and reached the scene at 02:54, the second responder arrived with the tool at 03:07, and the driver was extricated at 03:14, 27 minutes after the collision. The post-incident review found the extrication delayed by approximately 9 minutes, measured from 02:58, when the paramedic's primary survey was complete and the extrication plan made, to the second responder's arrival, and recorded that the delay did not affect the clinical outcome because the injuries were stable and the patient was conscious throughout. Argument Against: Nil Reasoning: The roster, the response log and the post-incident review agree on the times.	
Influence:	Fail, unlikely (10%)
Argument For: A nine minute wait for the second half of an extrication capability is a defence that is not there when it is needed, and nothing about the arrangement changes with the severity of the casualty it is called to. Argument Against: Nil Reasoning: The collision, the injuries and their severity were complete before the emergency response was paged. There is no path by which the response time affected this occurrence or its outcome, and the post-incident review says so on clinical grounds.	
Importance:	Pass, very likely (90%)
Argument For: It did not influence this occurrence and it is still a finding. The driver was conscious, stable, restrained and protected by an intact roll-over structure, and the delay cost him nothing. Change one thing, a fuel fire, an unconscious casualty, an airway that needed the vehicle opened in the first ten minutes, and the same nine minutes is a different outcome. The gap that matters is structural rather than clinical: the site's extrication capability on night shift is a single person plus a drive from the village, and no part of the arrangement scales with what has happened. Argument Against: Nil Reasoning: A known, structural gap in a last-line defence, on every night shift, that the site's own review has already measured at nine minutes.	

Local Conditions

Ramp surface wet or slippery at and above the intersection

Existence:	Fail, very unlikely (5%)
Argument For: Hypothesis tested because an 18 m push down a 9 per cent grade invites a question about the surface. It fails. The met station records no precipitation on the night and the last rainfall at the site was 26 days before the occurrence. The water cart log puts the last pass on that section at 01:10 (17:10 UTC), 1 hour 37 minutes before, and the standing night shift instruction is a pass every 90 minutes, so the surface had been drying for longer than the interval rather than freshly watered. The investigator examined the ramp at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection. The braked tyre scuff runs continuously for 58 m with no sign of a locked or sliding wheel, and the download gives a mean deceleration of 0.78 m/s² net, against the unit's own tested 0.80 and 0.82 on the same dry grade. There is no evidence of a wet or slippery surface from any source. Argument Against: Nil Reasoning: Three independent sources agree: the weather record, the surface as examined, and the deceleration actually achieved, which is the figure the truck achieves dry.	
Influence:	Not tested
Importance:	Not tested

Sight line of 41 m at the spur against a 96 m stopping distance

Existence:	Pass, virtually certain (100%)
Argument For: The survey pickup of 13/08/2026 places the spur mouth 34 m below a convex grade break in the ramp, where the descent steepens, which the crews call the crest. An operator descending at an eye height of 5.80 m first has line of sight to the spur mouth at 41 m, and to the top of a 4.5 m beacon whip on a light vehicle approaching the mouth at 53 m; a driver at the spur mouth at an eye height of 1.45 m first has line of sight to a descending truck at 38 m, with the 1.6 m inner windrow taking the truck's lower body out of view for the last 25 m of the spur's approach. The stopping distance of a 220 t class truck empty at the posted 40 km/h on a 9 per cent descending grade, on the manufacturer's data with a 1.5 second response, is 96 m. BIN-TMP-004 clause 6.2 requires available sight distance at every intersection to be not less than that stopping distance and in no case less than 120 m. Seven days earlier a dewatering technician reported the same sight line from the same position in his own words. Argument Against: Nil Reasoning: Surveyed to the eye heights of both vehicles by the mine survey department, and independently described by the hazard report seven days earlier.	
Influence:	Pass, very likely (90%)
Argument For: 41 m against a 96 m stopping distance means that a truck at the posted speed cannot stop for anything at this intersection, whatever the operator does and however alert he is. Argument Against: Nil Reasoning: The geometry put the first sight of the light vehicle's beacon at 53 m and of the vehicle itself at 41 m, against a truck needing 96 m to stop from the posted speed, and about 68 m on brakes alone from 37 km/h at the recorded 0.78 m/s². With twice the sight line, the same 1.4 second response and the same braking stop the truck short of the spur. The 10% allows that a sight line is only fatal once something is in it, and the other objects on the map are what put something in it.	
Importance:	Not tested

Airborne dust on the ramp above the intersection during the truck's approach

Existence:	Pass, likely (70%)
Argument For: The water cart's last pass on that section was 1 hour 37 minutes before the occurrence against a standing 90 minute interval, so a pass was 7 minutes overdue. The fixed dust monitor on the RL 328 bench beside the ramp, 180 m up-ramp of the intersection and above the grade break the truck came over, logged PM10 at 142 micrograms per cubic metre in the ten minutes covering the occurrence, against a shift mean of 88. The truck operator described dust hanging on the ramp, said the cart had been through earlier and it dries out, and the descent was behind his own lights into an unlit pit, which is the condition in which airborne dust is worst for a driver. Argument Against: The reading of 142 is well below the site trigger of 250 and is a measure of particulate at a monitor head, not of what could be seen from either cab; nothing at Bindarra measures visibility. The light vehicle driver was firm that the air was clear, that he could see the ramp surface and the windrow, and that the dust at Bindarra is nothing like the other sites he works at. Reasoning: Accepted at 70%. The deviation is real, in the right place, above the grade break the truck came over, and at the right time, at half as much again as the shift mean with the cart pass overdue. The driver's clear air is an honest account, but of the spur below the windrow, slow or stopped, not of the ramp above the crest where the truck was, which is why the dust is likely rather than certain.	
Influence:	Inconclusive, about as likely as not (40%)
Argument For: Dust hanging in the truck's lights over the crest is the condition in which a driver picks out a beacon late, and the operator described dust hanging around on the descent. Argument Against: The beacon was in view from 53 m, the first point at which the survey says it could be seen at all, in perfectly clear air, and the download puts the truck there at its last steady data point. A sighting at the geometric limit leaves dust nothing to explain. The operator himself declined to say it made a difference, and a 1.4 second response from that point, inside the plan's own allowance, leaves no room for a delay to hide in. Reasoning: Inconclusive at 40%, and recorded as inconclusive rather than failed. The case against is strong. What stops it failing outright is that the sighting is inferred from the last steady data point and the survey, not recorded, so the record cannot exclude dust costing a fraction of a second at the edge of the lights. The report owes the reader this reasoning rather than silence.	
Importance:	Not tested

Haul truck operator fatigued during the approach to the intersection

Existence:	Inconclusive, about as likely as not (50%)
Argument For: The operator was on night 10 of a 14 night swing, 8 hours 47 minutes into a 12 hour 30 minute shift, at 02:47 and inside the window of lowest circadian alertness. His own voluntary sleep record gives between 5 and 7 hours on each of the ten days before, 6 hours 30 minutes on the last of them, which is a short recovery repeated across the swing. He described himself as tired 'the way you are on night ten'. The task on the approach was detection in an unlit pit, which is the task fatigue degrades first. Argument Against: He described the tiredness as ordinary: 'Not falling asleep tired ... I was awake and I was driving properly.' Every limit in the contractor's fatigue standard was met. The recorded performance shows none of the usual signs: 1.4 seconds from the beacon coming into view to a brake pressure rise, inside the 1.5 seconds the plan's stopping distance assumes, and braking and steering that were correct and immediate. Reasoning: Inconclusive at 50%, and left there rather than dropped or promoted. 'Tired' on a tenth night is what he said, and it is not the same as fatigued to a degree that degrades performance; the recorded response argues against that and no other evidence reaches it. Settling it would need a detailed fatigue assessment, which was not pursued: even had Existence passed, Influence would fail, because braking at 0.78 m/s² from 53 m a response time of zero still ends in a collision, at about 17 km/h instead of 24.	
Influence:	Not tested
Importance:	Not tested

Individual Actions

Neither driver made a positive call before entering the intersection

Existence:	Pass, virtually certain (100%)
Argument For: The pit channel was logged continuously and the investigator listened to the whole period from the shift's return from crib to two minutes after the collision. There is no transmission by either vehicle about the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit. HT-407 called at each listed intersection it passed and was acknowledged: the RL 360 crossover going up at 02:28 and coming down at 02:46:21, and the ROM pad turn-off at 02:37. LV-118 made no transmission on the channel at all that shift beyond logging on at 18:04 (10:04 UTC), at the start of shift. No call was required: BIN-TMP-004 clause 4.3.2 requires none at an intersection not listed at Appendix B, and this one is not listed. The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently and without being led: calls are made at the nine intersections on the crib room board and at no others, and the supervisor said plainly that he has never corrected anybody for not calling elsewhere and that requiring calls everywhere would make the channel unusable. Argument Against: Nil Reasoning: Established by a continuous recording of the whole period and confirmed by all three interviews.	
Influence:	Pass, likely (85%)
Argument For: The one action that would have put each vehicle in the other's knowledge was available, cost nothing, and was on neither driver's mind, because no document put it there and the local practice matched the document exactly: the same truck had called at the crossover 51 seconds earlier. Argument Against: Nil Reasoning: A call from either vehicle would have been heard by the other, by dispatch and by the supervisor, and the dispatcher was already tracking the truck closely enough to place it. Discounted from certainty because a call gives knowledge, not separation, and the two vehicles still had to resolve who went first over 41 m of sight line.	
Importance:	Not tested

Light vehicle drove out of the spur onto the ramp without stopping

Existence:	Pass, virtually certain (100%)
Argument For: The driver's account, corrected by him on read-back from stopped to slowed, is that he slowed at the mouth, looked right up the ramp, saw nothing and drove out, turning right to go back up the ramp; he puts his speed coming out of the spur at about 25 km/h and does not believe he came to a complete stop. The collision awareness log places him at the spur mouth at 02:47:09.2, with the truck 21.4 m above it, and the impact is 6.2 m into the ramp from the spur mouth 2.8 seconds later. From the mouth, at his eye height, the survey gives 38 m of sight line to a descending truck, and the windrow takes the truck's lower body out of view for the last 25 m of the spur's approach. On the recorded times his look came a second or two before the truck came over the crest, when it was beyond anything his position let him see; the truck was within his 38 m from about 02:47:07.4, nearly two seconds before he reached the mouth. BIN-TMP-004 clause 4.1 requires light vehicles to give way to heavy vehicles at all times, which is the rule he described himself as applying. There was no stop or give-way control, no stop line, no warning sign and no posted speed at the mouth. He had done no Bindarra pit driving assessment and had never seen the traffic management plan. Argument Against: Nil Reasoning: His own account, volunteered and then corrected against his own interest, agrees with the collision awareness log and the scene measurements.	
Influence:	Pass, extremely likely (95%)
Argument For: This is the action that put a light vehicle across a descending haul truck's path, and it is the one place on the map where a different decision in the last ten seconds would have avoided the collision. Argument Against: Nil Reasoning: Without the light vehicle on the ramp there is no collision, and driving out without stopping is the proximate step. Had he stopped at the mouth and looked again, the truck was already inside his view from there, and he would very likely have seen it and let it pass, as the technician did by reversing on 04/08. The 5% allows that a stop is only as good as the look from it, at 38 m.	
Importance:	Not tested

Truck operator braked and steered right within 1.4 seconds of the beacon appearing

Existence:	Pass, virtually certain (100%)
Argument For: The download records the last steady data point at 02:47:06.0 (18:47:06.0 UTC) at 37 km/h, 53 m above the spur mouth, which is where the survey puts the first sight of a light vehicle's beacon over the crest, and service brake pressure rising from 0 to 620 kPa in 0.4 seconds from 02:47:07.4, 38.5 m above the mouth, with the retarder taken to maximum and 14 degrees of right steering input in the same movement. That is a 1.4 second response, inside the 1.5 seconds the plan's own stopping distance calculation allows. Mean deceleration from the brake application to the collision was 0.78 m/s² net, against the unit's tested 0.80. Impact speed was 24 km/h, six seconds after the beacon came into view; with no braking it would have been 37 km/h. The scene shows the braked scuff deviating 1.4 m right over its first 20 m, away from the open side of the road and towards the wall; to his left was the pit edge and a drop of more than 100 m. He stopped 19 m below the intersection, called the collision, followed the dispatcher's instruction to stay in the cab until the ramp was stopped, then went down and stayed with the driver until the response team arrived. Argument Against: Nil Reasoning: Recorded data throughout, corroborated by the scene marks.	
Influence:	Pass, very likely (90%)
Argument For: The difference between a 37 km/h and a 24 km/h impact on the side of a light vehicle is the difference between the injuries actually sustained and a very different outcome. Steering towards the wall kept the push running down the ramp, so the vehicle came to rest against the pit side windrow rather than being driven through it and over the edge. Argument Against: Nil Reasoning: Recorded as a decrease in risk. Braking from 37 to 24 km/h in 4.6 seconds took about 58% of the impact energy out, since $(24/37)^2$ is 0.42, and the steering input turned the truck away from the open edge.	
Importance:	Not tested

Occurrence Events

Service brakes delivered less than their tested deceleration during the stop

Existence:	Fail, very unlikely (5%)
Argument For: Hypothesis tested because a 220 tonne truck that does not stop in the distance available raises the brakes first. It fails. The monthly brake performance test on 03/08/2026, eight days before, gave a mean deceleration of 0.80 m/s² net in the empty condition on a 9 per cent descending grade. The test repeated on 14/08/2026 before the unit returned to service gave 0.82. The download of the occurrence itself gives 0.78 over the 4.6 seconds of braking, 37 km/h to 24 km/h, which is inside the normal scatter of the two tests. Service brake pressure rose from 0 to 620 kPa in 0.4 seconds, the retarder was at maximum throughout, and no brake, retarder or park brake defect was raised against the unit in the twelve months before the occurrence. The pre-start check sheet for 11/08/2026 is complete and records no defects. The braking system performed to specification before, during and after the occurrence. Argument Against: Nil Reasoning: Two performance tests either side of the occurrence and the occurrence download itself agree to within 5 per cent, and no defect history exists.	
Influence:	Not tested
Importance:	Not tested

Neither driver aware of the other until the truck came over the crest

Existence:	Pass, extremely likely (95%)
Argument For: HT-407 was descending the ramp empty at 37 km/h with no intersection shown on its in-cab road network, no warning sign on the approach, no call on the channel and no alert from a collision awareness system that later logged the light vehicle at 21.4 m. LV-118 was coming out of the spur towards a mouth with 38 m of sight line, no give-way control, no call and no knowledge of the truck. The first sighting available to the truck operator was the light vehicle's beacon over the crest at 53 m, six seconds before the impact, which is what the survey predicts for the geometry; the light vehicle driver's look up the ramp came before the truck was within his view. Neither the radio log, the download, the CAS log nor any of the three interviews contains a single indication that either driver knew the other was there before that moment. The night shift supervisor did not know a light vehicle was in the pit at all, and there is no permit, call-in or record by which he could have. Argument Against: Nil Reasoning: Established by three independent recordings and all three interviews. The 5% reflects that the light vehicle's movements before it reached the spur mouth rest on the driver's own account alone.	
Influence:	Pass, extremely likely (95%)
Argument For: Every organisational and control finding on this map converges on this one condition: two vehicles in the same place, at night, each unknown to the other until six seconds remained. Argument Against: Nil Reasoning: The collision follows directly and immediately from the convergence; nothing between the two could have prevented it in the six seconds left.	
Importance:	Not tested

The haul truck's front left struck the right side of the light vehicle at 24 km/h

Existence:	Pass, virtually certain (100%)
Argument For: Impact at 02:47:12.0 (18:47:12.0 UTC) at 24 km/h, 4.6 seconds after the brake application and 6.2 m into the ramp from the spur mouth, 3.1 m out from the toe of the inner (pit side) windrow. LV-118 was crossing the descending lane, turning right to go up the ramp, with its right side to the truck. The truck's accelerometer logged 1.9 g longitudinal. Damage to LV-118 is concentrated on the right side from the B pillar forward; damage to HT-407 is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line. Right side glass and mirror fragments lie from 4.8 m to 14 m down-ramp of the impact position. Argument Against: Nil Reasoning: Recorded data, the damage pattern and the debris field all agree.	
Influence:	Pass, virtually certain (100%)
Argument For: A 220 tonne vehicle striking the side of a 3 tonne one leaves the outcome entirely to the light vehicle's structure, the restraint and the speed, and two of those three were already settled before the sighting. Argument Against: Nil Reasoning: The impact is the mechanism by which the convergence became an injury.	
Importance:	Not tested

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

Existence:	Pass, virtually certain (100%)
Argument For: A continuous gouge runs 18.4 m down-ramp from the impact position, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over. LV-118 came to rest on its left side against the inner windrow, facing down-ramp, 18.4 m below the impact. The roll-over protection structure is deformed inward at the right rear mounting and was not breached. The driver's seatbelt was found fastened with loading marks on the webbing. The occupant sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout. The door on the resting side was against the ground, so extrication was through the rear of the vehicle. Argument Against: Nil Reasoning: Scene measurements, the vehicle examination and the medical record agree.	
Influence:	Pass, virtually certain (100%)
Argument For: The injuries are the ones a side impact and a roll-over produce when the structure holds and the restraint is worn. The same sequence with either of those absent is a different report. Argument Against: Nil Reasoning: The roll-over is how the collision produced the injuries that make this a serious occurrence.	
Importance:	Not tested

Light vehicle driver seriously injured in a collision with a haul truck on the main ramp

Existence:	Pass, virtually certain (100%)
Argument For: At 02:47:12 on 12/08/2026 (18:47:12 UTC on 11/08/2026) a 220 t class rigid haul truck descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine empty struck the right side of a light vehicle coming out of the RL 312 dewatering spur at 24 km/h, after braking from 37 km/h, pushed it 18.4 m down the ramp and rolled it onto its left side against the inner windrow. The light vehicle carried one occupant, a grade control technician employed by a survey subcontractor, on his third night shift at the mine. He was conscious throughout, was extricated through the rear of the vehicle 27 minutes after the collision, and was flown to the metropolitan trauma centre at 07:40. Argument Against: Nil Reasoning: Nil	
Influence:	Not tested
Argument For: A serious injury: a fractured left clavicle, three fractured left ribs and a small pneumothorax, treated at the surface medical facility and evacuated by air. The light vehicle was destroyed; its roll-over protection structure was deformed inward at the right rear mounting and the seatbelt webbing carried loading marks, so two defences were loaded to the point of showing it. The haul truck was damaged at the front left and returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury. Argument Against: Nil Reasoning: Nil	
Importance:	Not tested